Worksheet 8.5

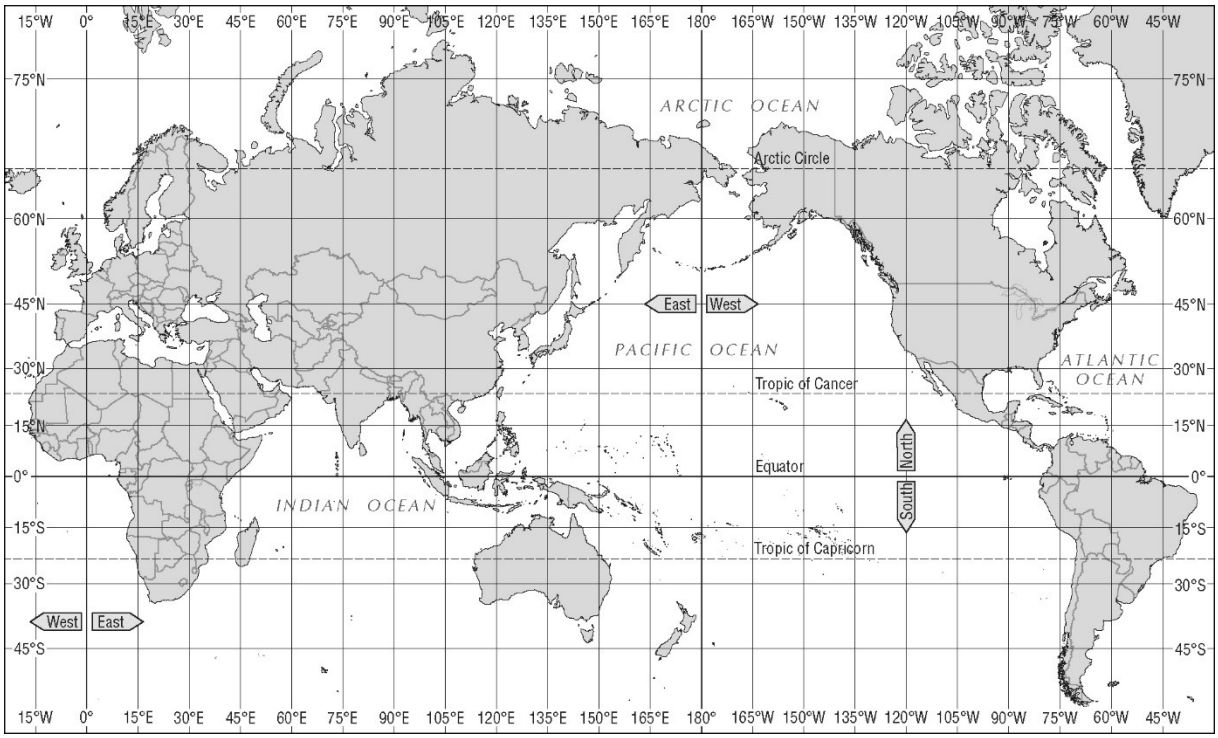
Plotting earthquake activity

Answers

The following table contains data for large earthquakes that have occurred in the Pacific Rim in the last 100 years.

	Magnitude	Location	Latitude	Longitude
A	8.5	Kuril islands	45°N	150°E
B	8.5	Argentina	29°S	71°W
C	8.5	Banda Sea	5°S	132°E
D	8.6	Tibet	29°N	96°E
E	8.6	Indonesia	2°N	97°E
F	8.8	Ecuador (off coast)	1°N	82°W
G	9.0	Kamchatka	53°N	160°E
H	9.1	Sumatra	3°N	96°E
I	9.2	Alaska	61°N	148°W
J	9.5	Chile	38°S	73°W

1. Plot the location of each earthquake on the map below.



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2. Account for the location of these earthquakes in relation to the tectonic plates.

Most of these earthquakes are along the Pacific Rim except for the Tibetan earthquake. The border of the Pacific represents the edges of the ocean plate and continental plates. These are zones of subduction, which lead to earthquakes and volcanic activity as the ocean plate is pushed below the continental plates. The Tibetan earthquake is the result of the Indo-Australian plate colliding with the Eurasian plate.

3. The Chilean earthquake occurred in 1960. What effect does an earthquake of this magnitude have on the ground surface?

An earthquake with a magnitude of 9.5 is catastrophic. The ground rises and falls during the event, resulting in complete devastation in the zone of the epicentre.

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